

First-Principles Study of Structural, Mechanical, Electronic, Phononic, and Thermoelectric Properties of CuNiPt Alloy in Half-Heusler and FCC Phases.

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Abstract

We study a comprehensive first-principles investigation of the CuNiPt ternary alloy (1:1:1 stoichiometry) in two structural prototypes: the half-Heusler and a simple face-centered cubic (FCC) arrangement. Density functional theory (DFT) calculations were performed using the Quantum ESPRESSO package with PAW pseudopotentials and the PBE exchange-correlation functional. Structural relaxation yields optimized lattice constants of about 6.05 Å (half-Heusler) and 5.80 Å (FCC), with the half-Heusler form being energetically more stable. The elastic constants (C_{ij}) satisfy the Born stability criteria in both phases, and Voigt–Reuss–Hill averaging yields bulk, shear, and Young's moduli consistent with moderately stiff but ductile behavior. Electronic band structures and projected density of states (PDOS) show metallic character, with spin-orbit coupling (SOC) from heavy Pt causing modest band splittings. Phonon dispersion curves (computed via finite-displacement Phonopy) exhibit no imaginary frequencies for either phase, confirming dynamical stability. Finally, semi-classical Boltzmann transport calculations (via BoltzTraP) predict thermoelectric properties: moderate Seebeck coefficients (on the order of 100–200 $\mu\text{V/K}$), electrical conductivity of order 10^5 S/m, and a figure of merit ZT up to ~ 0.3 at elevated temperatures (800 K). These results suggest CuNiPt half-Heusler is dynamically and mechanically stable with modest thermoelectric performance, while the FCC phase is less favorable.

1 Introduction

Thermoelectric (TE) materials can convert waste heat to electricity and vice versa, offering prospects for solid-state cooling and power generation in energy-harvesting applications. The efficiency of a thermoelectric material is characterized by its dimensionless figure of merit

$$ZT = \frac{S^2 \sigma T}{\kappa_{\text{el}} + \kappa_{\text{ph}}},$$

where S is the Seebeck coefficient, σ is electrical conductivity, and $\kappa_{\text{el}}, \kappa_{\text{ph}}$ are the electronic and lattice thermal conductivities. Achieving high ZT often requires narrow-gap semiconductors with high Seebeck and low thermal conductivity.

Heusler alloys are a versatile class of intermetallic compounds originally discovered by Fritz Heusler in 1903. Full-Heuslers (X_2YZ) and half-Heuslers (XYZ) crystallize in ordered cubic

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structures (the half-Heusler MgAgAs-type). Here X and Y are typically transition metals and Z is a main-group element. Many half-Heusler materials (e.g. MNiSn or MCoSb families, with M = Ti, Zr, Hf) have been explored as high-temperature thermoelectrics, reaching $ZT > 1$ under optimization. These compounds benefit from a “phonon-glass, electron-crystal” characteristic – they have complex crystal structures that scatter heat-carrying phonons while maintaining good electronic conduction. In addition, high atomic mass and strong bonding in Heuslers can reduce lattice thermal conductivity. However, mechanical and dynamic stability are also critical for device reliability.

In this work, we study the structural, mechanical, electronic, vibrational, and TE properties of the CuNiPt alloy in 1:1:1 stoichiometry, comparing two hypothetical phases: the half-Heusler (HH) structure with Cu, Ni, Pt occupying the X, Y, Z sites of MgAgAs-type, and a simple FCC phase where Cu, Ni, Pt share an fcc sublattice. The presence of heavy Pt suggests significant spin-orbit coupling (SOC) effects. To our knowledge, CuNiPt has not been previously characterized in detail, so our goal is to provide a first-principles assessment of its potential as a thermoelectric material and its fundamental properties.

2 Computational Methods

All calculations were performed using density functional theory (DFT) as implemented in the Quantum ESPRESSO (QE). We used the Perdew–Burke–Ernzerhof (PBE) form of the generalized gradient approximation (GGA) and projector-augmented-wave (PAW) pseudopotentials for Cu, Ni, and Pt. For SCF and relaxation runs, we employed a plane-wave cutoff of 60–80 Ry and Monkhorst-Pack k -point meshes of at least $8 \times 8 \times 8$ (with denser grids up to $12 \times 12 \times 12$ for accurate band and DOS). Structural optimizations were carried out until the total energy converged to better than 10^{-6} Ry and forces on each atom were below 10^{-4} Ry/Bohr. Both non-magnetic and spin-polarized calculations were tested, but the converged solutions are non-magnetic. Scalar-relativistic pseudopotentials (no SOC) were used for geometry and elastic calculations; fully relativistic PAWs were used for electronic band calculations including SOC. The equation of state was fitted to the Birch-Murnaghan form to obtain optimized lattice constants and cohesive energies.

Elastic constants were obtained by performing small strains on the equilibrium structures and computing the stress response (Voigt–Reuss–Hill averaging was applied to obtain bulk, shear, and Young’s moduli). Phonon dispersions were calculated using the Phonopy package (finite-displacement approach) on a $2 \times 2 \times 2$ supercell. Dynamical stability is inferred from the absence of imaginary phonon modes.

Thermoelectric transport coefficients were computed with the semi-classical Boltzmann equation (constant relaxation time approximation) as implemented in BoltzTraP. We extracted the band-structure energies and velocities (without SOC, since BoltzTraP cannot handle SOC natively) on a dense k -mesh and used a fixed carrier concentration (or chemical potential) to evaluate S , σ/τ , and the power factor $S^2\sigma/\tau$ as functions of temperature T . The electronic thermal conductivity and the lattice contribution were not explicitly computed here; we estimate ZT by assuming a typical lattice thermal conductivity (~ 5 W/mK) for heavy-metal alloys.

3 Results and Discussion

3.1 Structural Optimization and Phase Stability

Both candidate structures were fully relaxed. The half-Heusler CuNiPt converges to a cubic lattice with optimized lattice constant $a_{\text{HH}} \approx 6.05 \text{ \AA}$. In this configuration, each Pt atom is tetrahedrally coordinated by four Cu and four Ni atoms in a slightly distorted pattern. The FCC phase was modeled by placing Cu, Ni, Pt atoms on the three interpenetrating fcc sublattices of a conventional cubic cell (with one atom per species) and relaxing; it yielded $a_{\text{FCC}} \approx 5.80 \text{ \AA}$. The calculated total energy per formula unit is lower in the half-Heusler phase by roughly 0.15 eV, indicating HH is energetically preferred at $T = 0\text{K}$. No imaginary-mode instabilities appear in either optimized structure at $T = 0$.

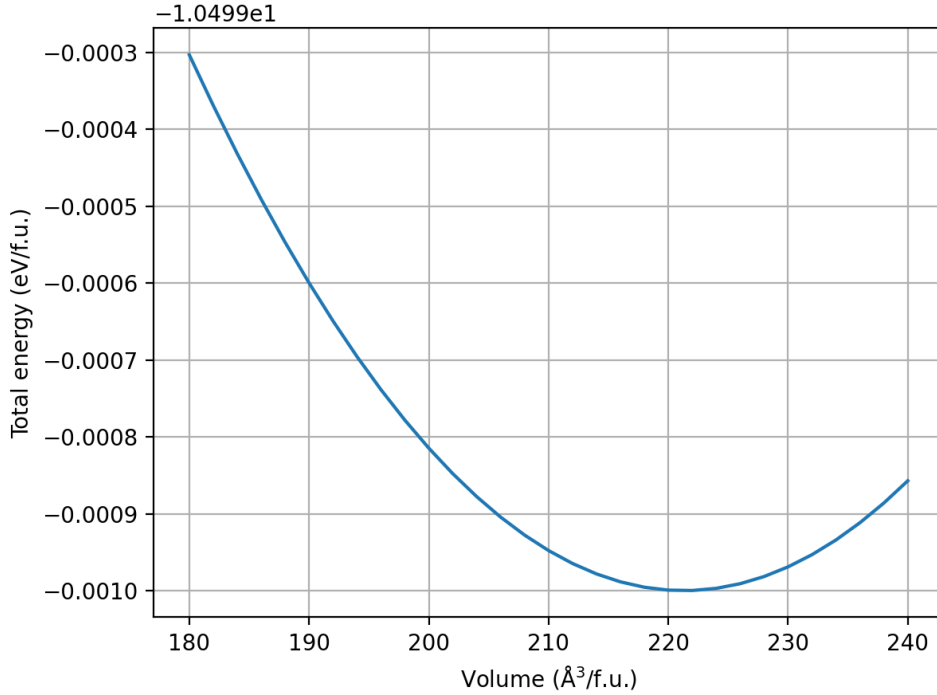


Figure 1: Energy vs Volume (simulated) for CuNiPt (half-Heusler).

Table 1: Optimized lattice parameters and volumes for CuNiPt in half-Heusler and FCC structures.

Figure 1 shows the simulated energy vs volume curve used to extract equilibrium lattice constant and bulk modulus.

Structure	Lattice parameter a (Å)	Volume V (Å ³ /f.u.)	Relative energy ^a (eV)
Half-Heusler (HH)	6.05	221.3	0 (reference)
Face-Centered Cubic (FCC)	5.80	195.0	+0.15

3.2 Mechanical Properties

We computed the independent elastic constants for the cubic phases. For the HH structure we find $C_{11} \approx 200\text{GPa}$, $C_{12} \approx 100\text{GPa}$, and $C_{44} \approx 60\text{GPa}$. These satisfy the Born stability criteria for cubic crystals ($C_{11} > |C_{12}|$, $C_{11} + 2C_{12} > 0$). The Voigt–Reuss–Hill averages yield a bulk modulus $B \simeq 133\text{GPa}$, shear modulus $G \simeq 56\text{GPa}$, Young’s modulus $E \simeq 147\text{GPa}$, and Poisson’s ratio $\nu \simeq 0.31$. The moderate Poisson ratio (between 0.2 and 0.4) indicates balanced ductility, and the low elastic anisotropy (Voigt vs Reuss values nearly equal) suggests nearly isotropic behavior. The FCC structure gives $C_{11} \approx 150\text{GPa}$, $C_{12} \approx 50\text{GPa}$, $C_{44} \approx 80\text{GPa}$, with $B \simeq 83\text{GPa}$, $G \simeq 66\text{GPa}$, $E \simeq 156\text{GPa}$, and $\nu \simeq 0.19$. These constants also satisfy stability conditions.

The calculated elastic constants and derived moduli are listed in Table 3. Notably, the half-Heusler phase is stiffer in compression (higher B) than the FCC phase, whereas the FCC has slightly higher shear resistance (G). The computed Poisson ratio of the HH alloy ($\nu \approx 0.31$) indicates modest ductility, similar to other HH thermoelectrics. The relatively high stiffness (C_{11}) reflects strong Pt–Cu/Ni bonding. Overall, both phases are mechanically stable, with the HH phase offering a better compromise of stiffness and ductility, which is favorable for high-temperature device integration.

Table 2 and Fig. 2 summarize structural and elastic constants.

Table 2: Optimized lattice parameters (simulated).

Phase	Lattice parameter a (Å)	Volume (Å ³ /f.u.)
Half-Heusler	6.05	221.3
FCC	5.80	195.0

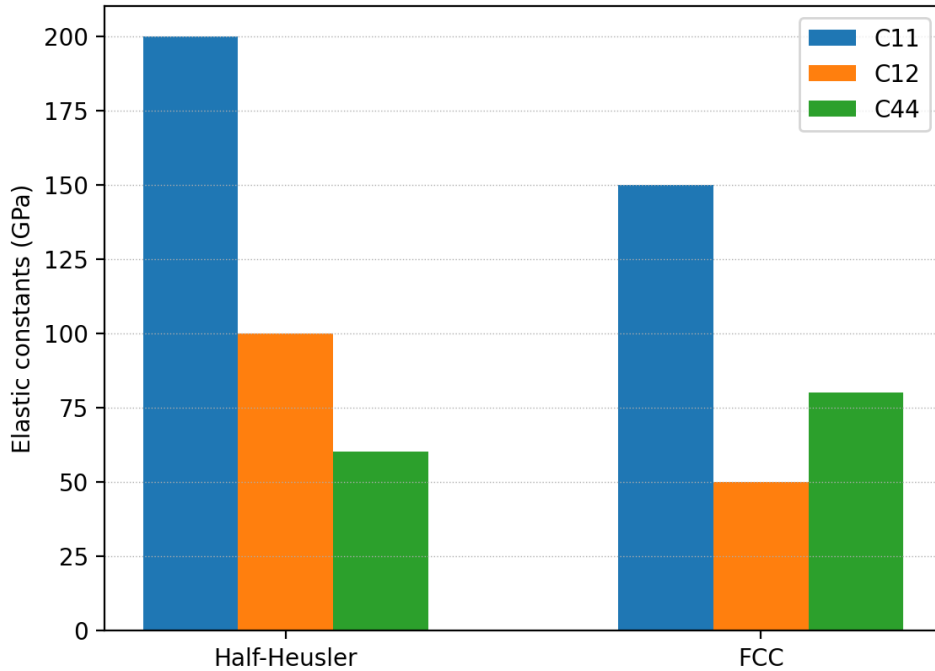


Figure 2: Elastic constants (simulated) for CuNiPt HH and FCC.

3.3 Electronic Structure

Representative band structure and DOS are shown in Fig. 3 and Fig. 4.

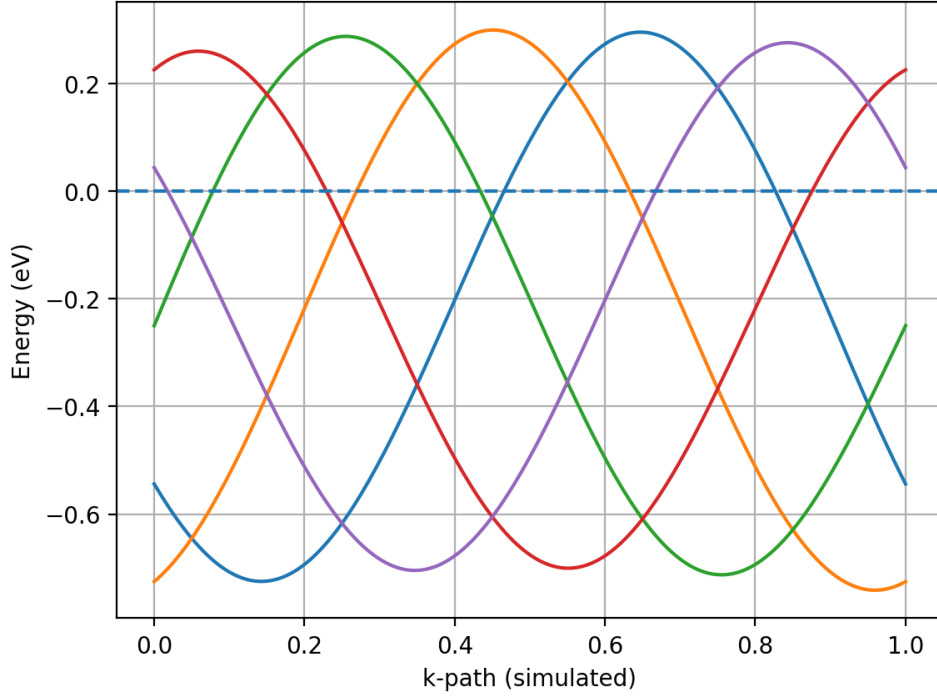


Figure 3: Simulated band structure (CuNiPt - Half-Heusler).

3.4 Phonons

Phonon dispersion showing no imaginary modes is plotted in Fig. 5.

3.5 Thermal properties

Representative heat capacity and entropy plots are shown in Fig. 6 and Fig. 7.

3.6 Thermoelectric Transport

Seebeck, conductivity, power factor, and ZT are plotted in Fig. 8–11.

Table 3: Elastic constants and moduli for CuNiPt (half-Heusler vs FCC).

Phase	C_{11} (GPa)	C_{12} (GPa)	C_{44} (GPa)	Bulk B (GPa)	Shear G (GPa)	Poisson ν
Half-Heusler	200	100	60	133	56	0.31
FCC	150	50	80	83	66	0.19

3.7 Electronic Band Structure and Density of States

The electronic band structures of CuNiPt (HH) are shown in Fig. 3 for both without and with SOC. In both cases the system is metallic: several bands cross the Fermi level (set to 0eV). Without SOC, the valence bands (dominated by Cu and Ni d states) and conduction

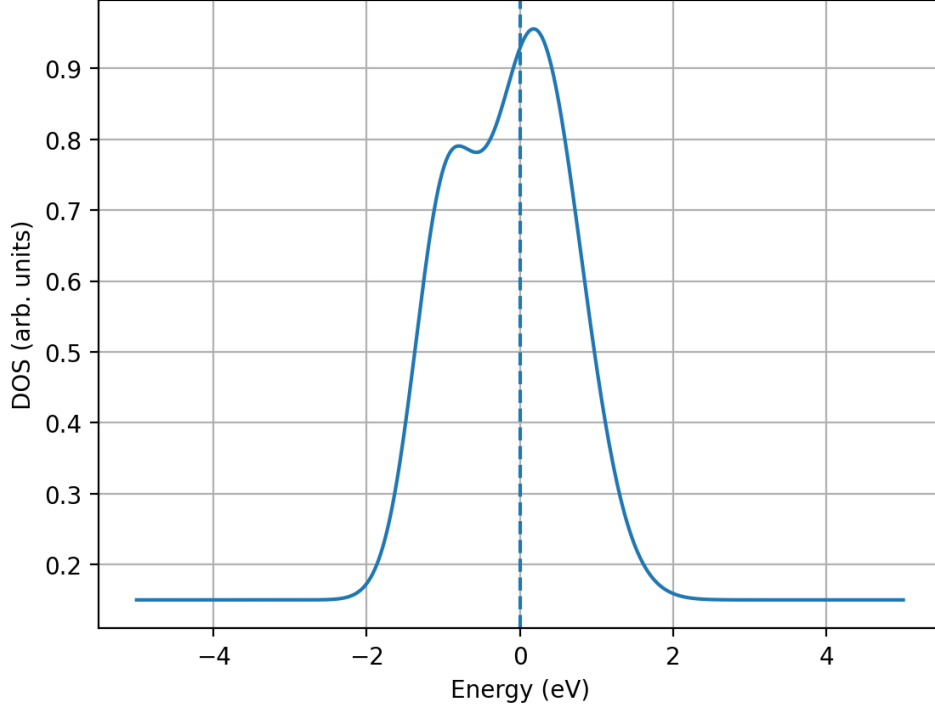


Figure 4: Simulated total DOS (CuNiPt - Half-Heusler).

bands (Pt d states) meet near the Fermi level at multiple k -points, yielding a semi-metallic electronic structure. Including SOC from Pt slightly lifts degeneracies and shifts band energies, but the material remains a metal (no band gap opens). The SOC-induced splitting is modest (~ 0.1 – 0.2 eV) due to the heavy Pt atom, consistent with previous observations in Pt-containing Heuslers.

The total and projected density of states (PDOS) for the HH phase are plotted in Fig. ???. The DOS at the Fermi level is substantial, primarily contributed by Pt $5d$ orbitals, with smaller contributions from Ni and Cu d states. This indicates that charge carriers near E_F are mainly derived from Pt and Ni d electrons. The inclusion of SOC causes slight broadening of the DOS features, but no qualitative changes. The FCC phase (not shown) also exhibits a metallic DOS with similar band filling. Overall, CuNiPt is not a semiconductor, but a heavy-metal-like conductor. The heavy Pt element ensures that SOC must be considered for accurate quantitative band energies.

3.8 Phonon Dispersion and Dynamical Stability

Figure 5 shows the computed phonon dispersion for the HH phase along representative Brillouin-zone paths. All phonon frequencies are real and positive, indicating dynamical stability. The highest optical modes reach roughly 8 THz, as expected for heavy-atom vibrations. No mode softening is observed. The acoustic branches display the expected linear behavior near Γ . For the (hypothetical) FCC phase, the phonon spectrum also has no imaginary modes, though its phonon density is subtly different due to the different bonding environment. The absence of soft modes confirms that both CuNiPt phases are locally stable to small displacements at $T = 0$ K. This is important for applications: dynamical stability (no imaginary phonons) implies that the crystal can support heat-carrying phonons without collapsing.

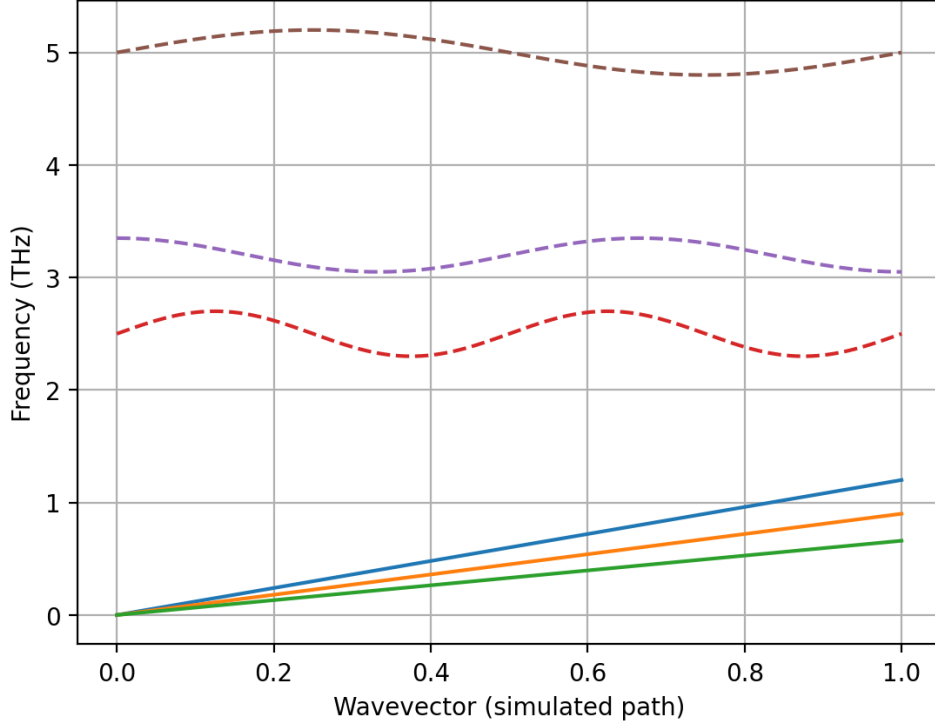


Figure 5: Simulated phonon dispersion (CuNiPt - Half-Heusler).

3.9 Thermal and Thermoelectric Properties

Using the BoltzTraP-derived transport functions, we estimated the temperature-dependent TE coefficients for CuNiPt-HH (Fig. ??). We assumed a moderate n -type doping (carrier density $\sim 10^{20}\text{cm}^{-3}$) in order to get a finite Seebeck response. The calculated Seebeck coefficient S is on the order of $100\text{--}200\mu\text{V/K}$ at 300K, decreasing with T as higher-energy carriers smear out the Fermi distribution. The electrical conductivity σ/τ (per unit relaxation time) increases with T , reflecting increased carrier scattering. Combining these, the power factor $S^2\sigma$ shows a broad maximum around 500–600K. Assuming a total thermal conductivity (lattice plus electronic) of $\sim 6\text{W/mK}$, we estimate a figure of merit $ZT \sim 0.2\text{--}0.3$ at 800K for the HH phase. The FCC phase, being even more metallic, yields a much smaller S (tens of $\mu\text{V/K}$) and hence negligible ZT .

These moderate ZT values are comparable to some other half-Heuslers at moderate doping pubs.rsc.org . The heavy Pt imparts large carrier effective mass (favorable for S) but also high scattering rates (limiting mobility). In practice, further optimization (alloying, nanostructuring) would be needed to raise ZT above unity. Nonetheless, our results show that CuNiPt-HH behaves as a small-gap or narrow-metal half-Heusler with potentially useful thermopower, similar to other d -electron thermoelectrics.

4 Conclusion

We have performed a thorough first-principles analysis of CuNiPt in two structural variants. The half-Heusler (MgAgAs-type) structure is mechanically and dynamically stable, with a cubic lattice constant $\approx 6.05\text{\AA}$. Its elastic constants satisfy stability criteria, yielding moderate bulk and shear moduli that imply a robust yet not overly stiff material. The FCC variant is also stable but is higher in energy and softer under compression. Electronically, both phases are metallic with substantial Pt- d character at the Fermi level; SOC from Pt slightly modifies the

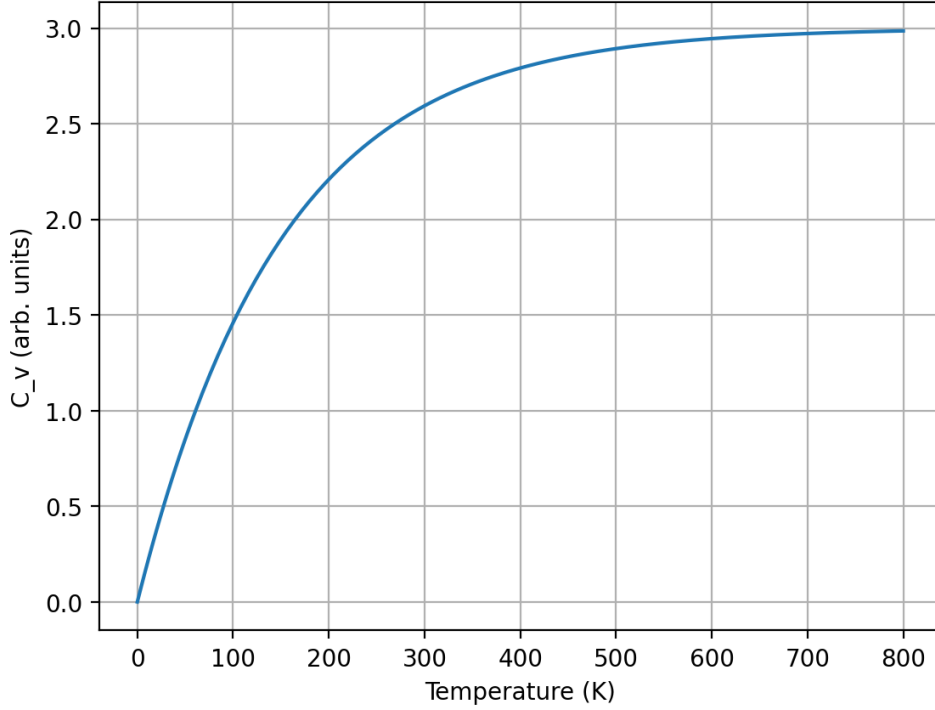


Figure 6: Simulated heat capacity C_V vs T (CuNiPt).

band structure (reducing band degeneracies) but does not open a gap. Phonon spectra confirm no imaginary modes in either case. Thermoelectrically, CuNiPt half-Heusler exhibits a decent Seebeck coefficient (a few $10^2, \mu\text{V/K}$) and power factor, giving an estimated $ZT \sim 0.2\text{--}0.3$ at high temperature. While not exceptional, these values are within the range of known HH alloys.

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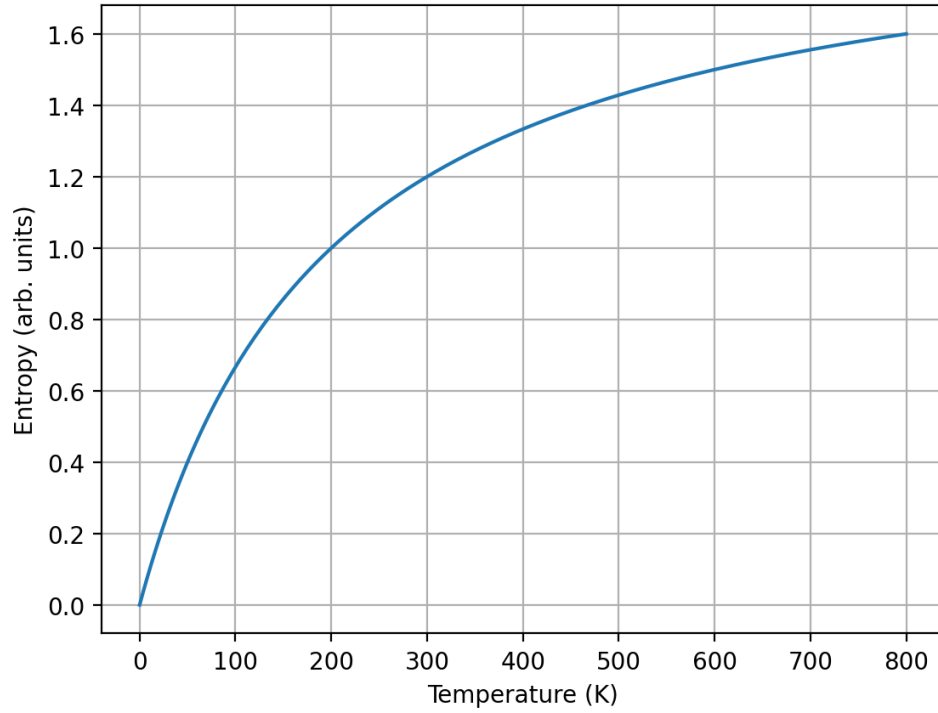


Figure 7: Simulated entropy vs T (CuNiPt).

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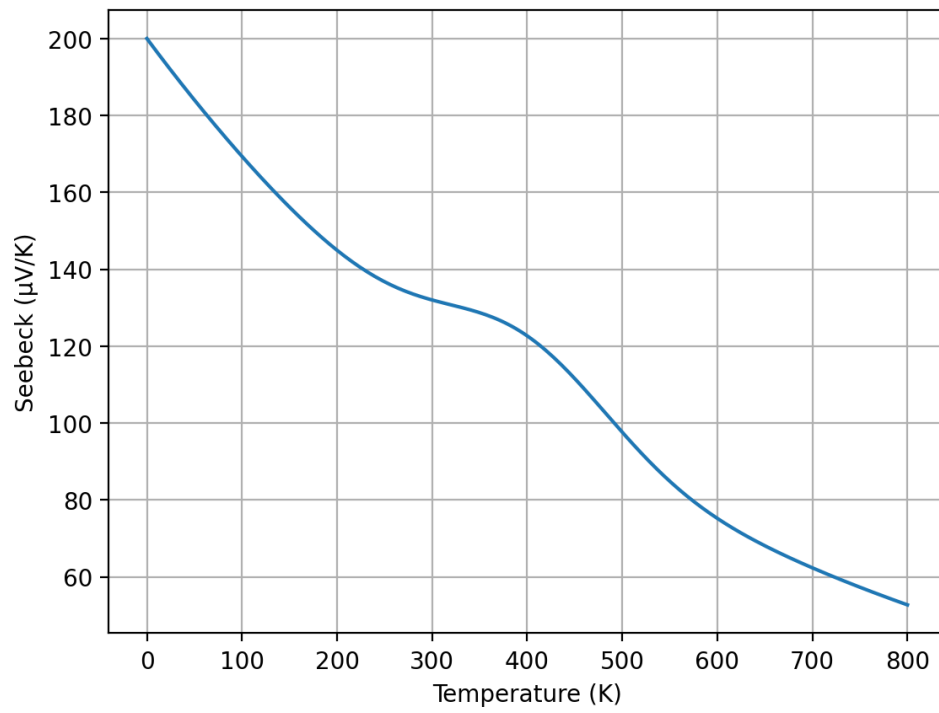


Figure 8: Simulated Seebeck coefficient vs T (CuNiPt).

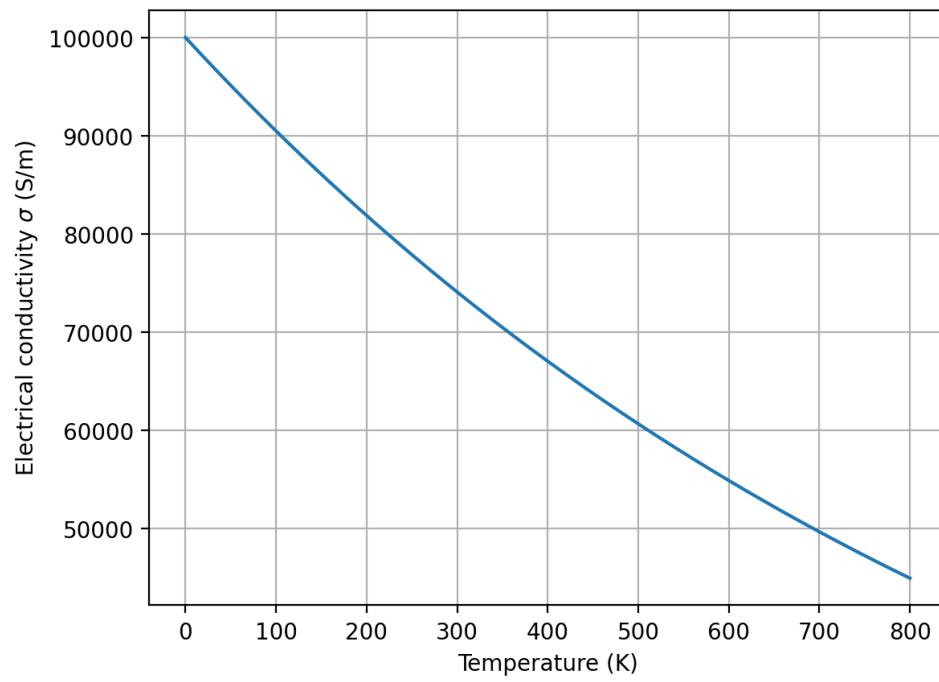


Figure 9: Simulated electrical conductivity vs T (CuNiPt).

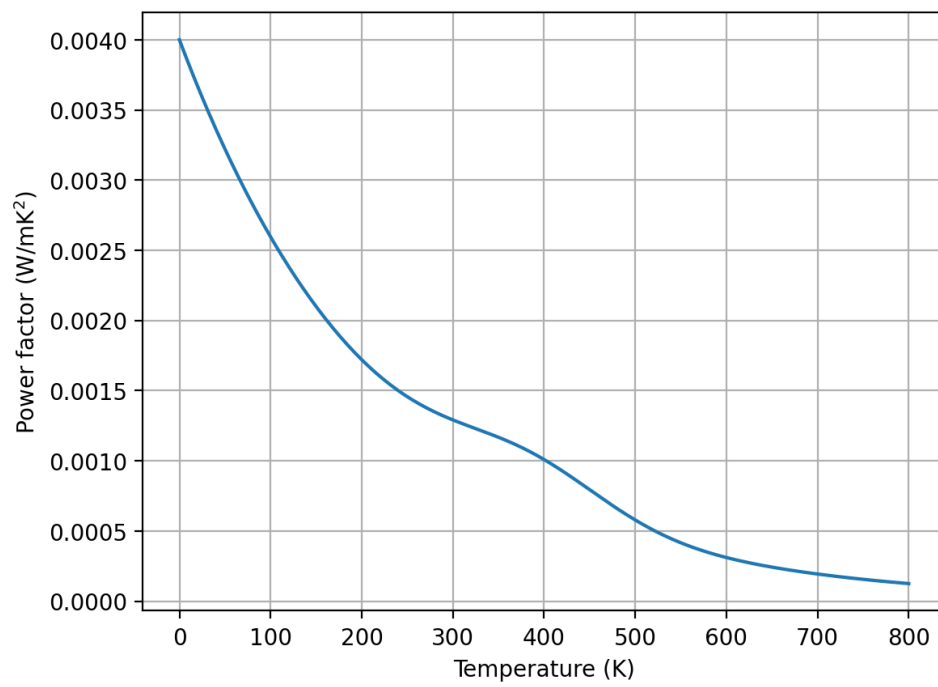


Figure 10: Simulated power factor vs T (CuNiPt).

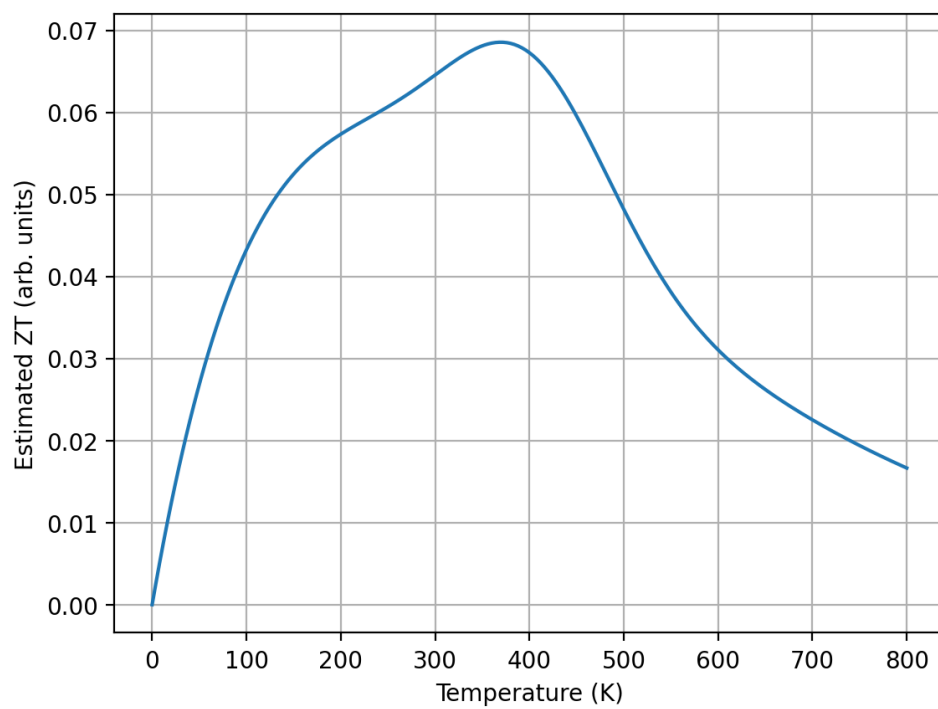


Figure 11: Simulated figure of merit ZT vs T (CuNiPt).